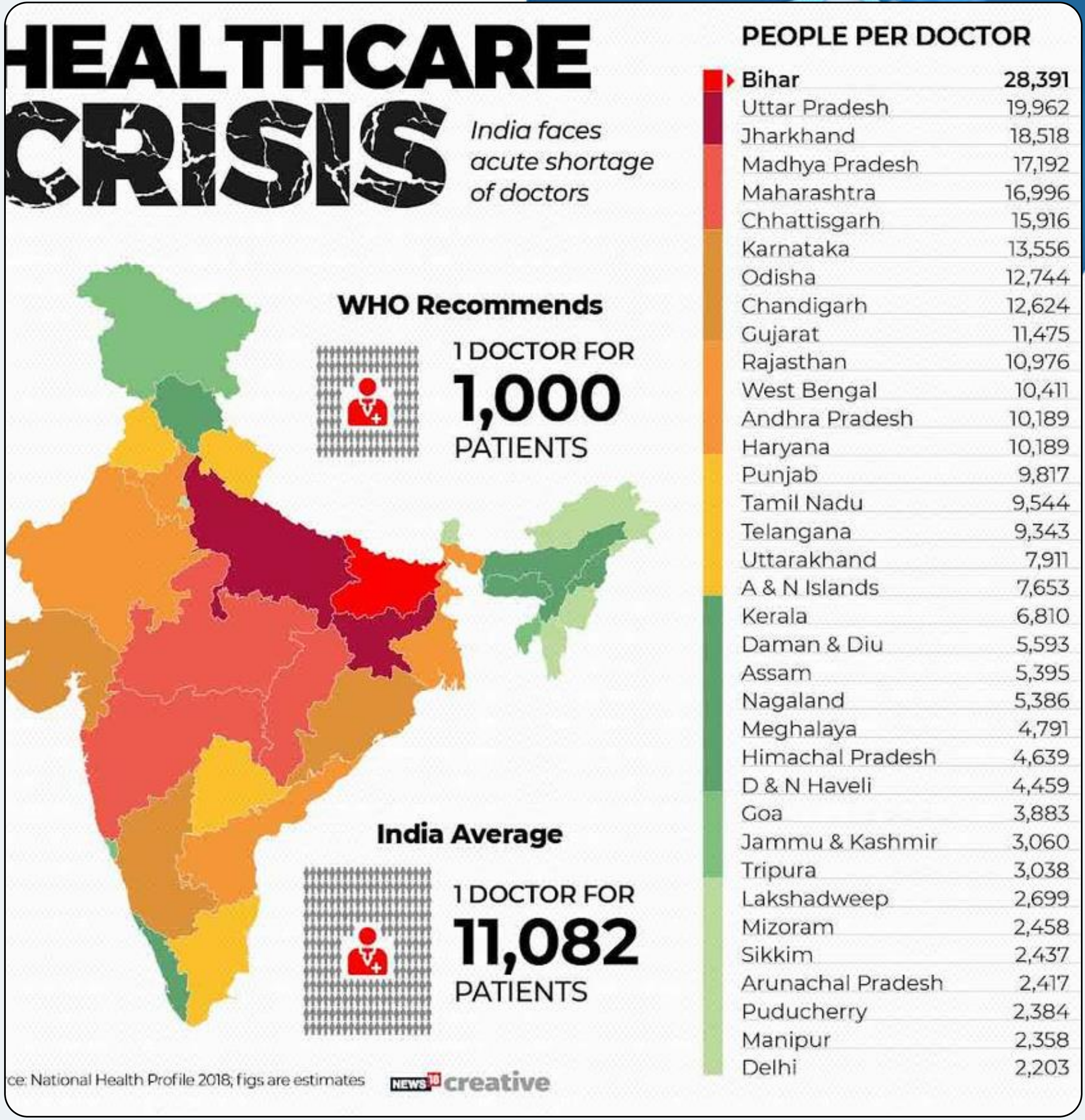
Intelligent User-Specific Healthcare Tracking

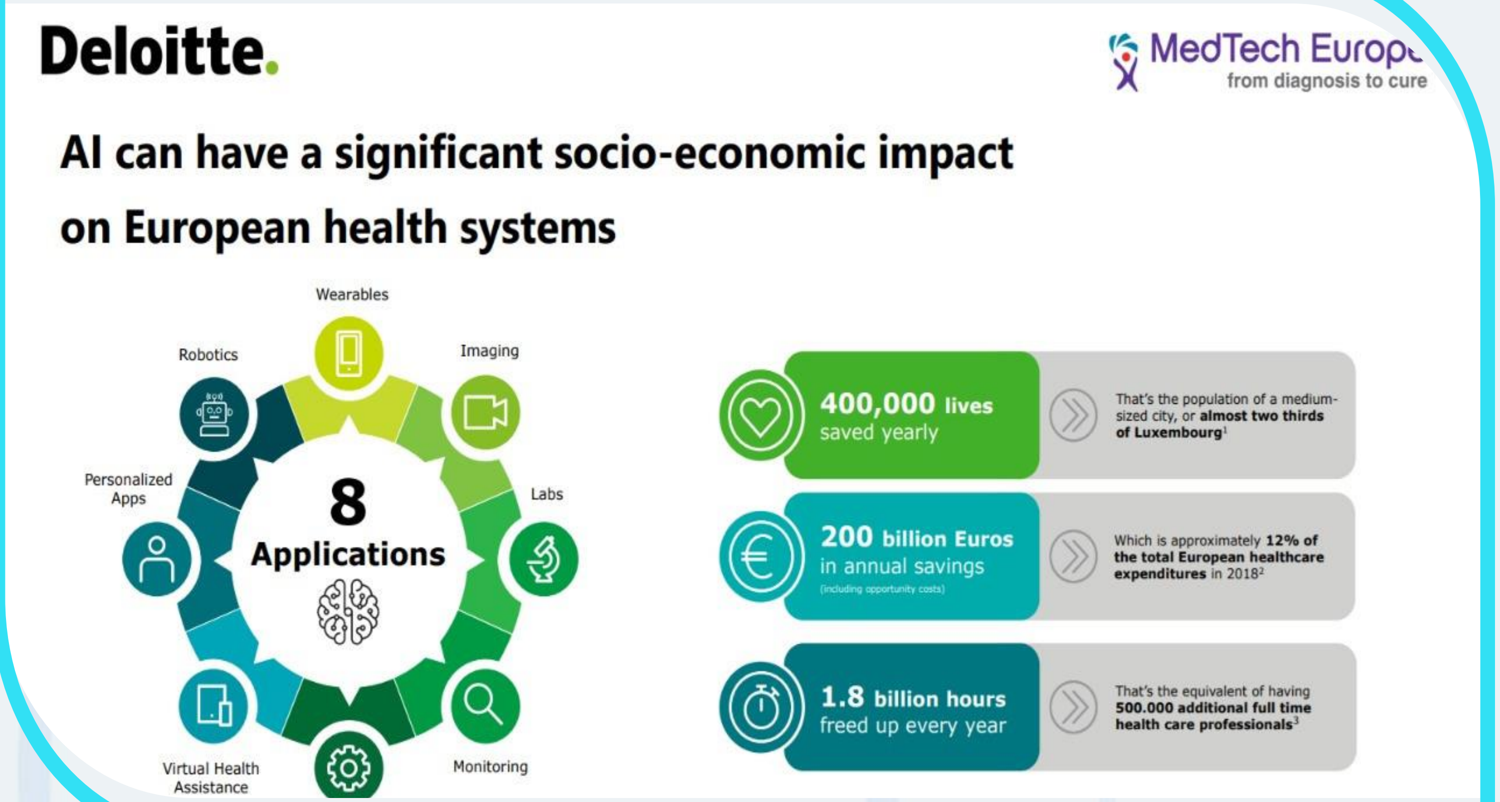
MedAI

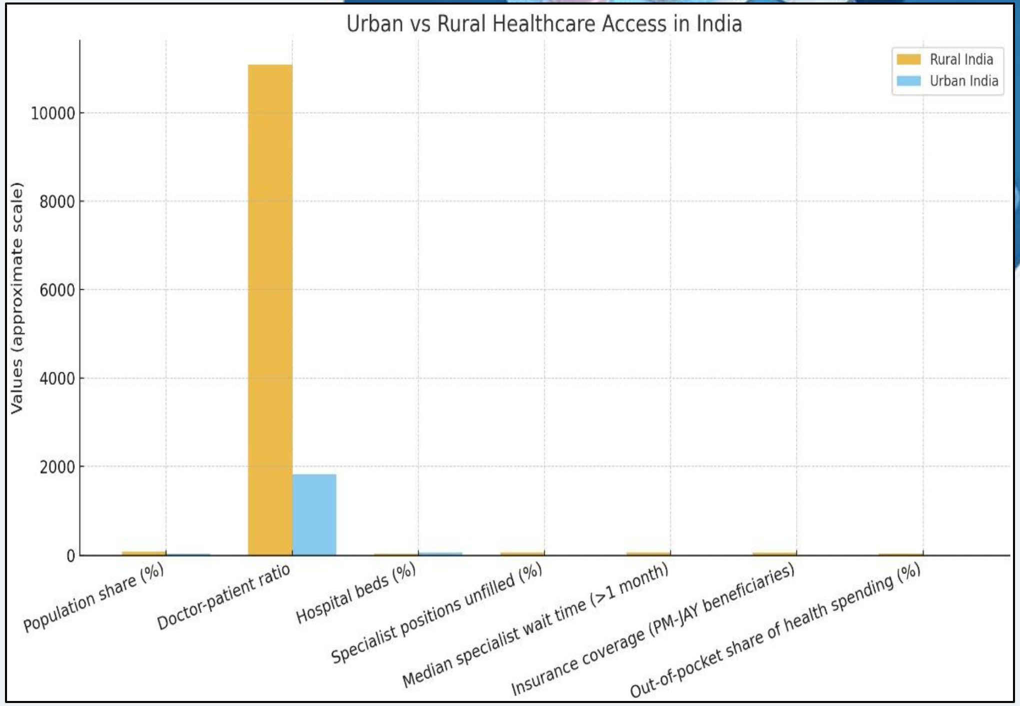
PROBLEM STATEMENT

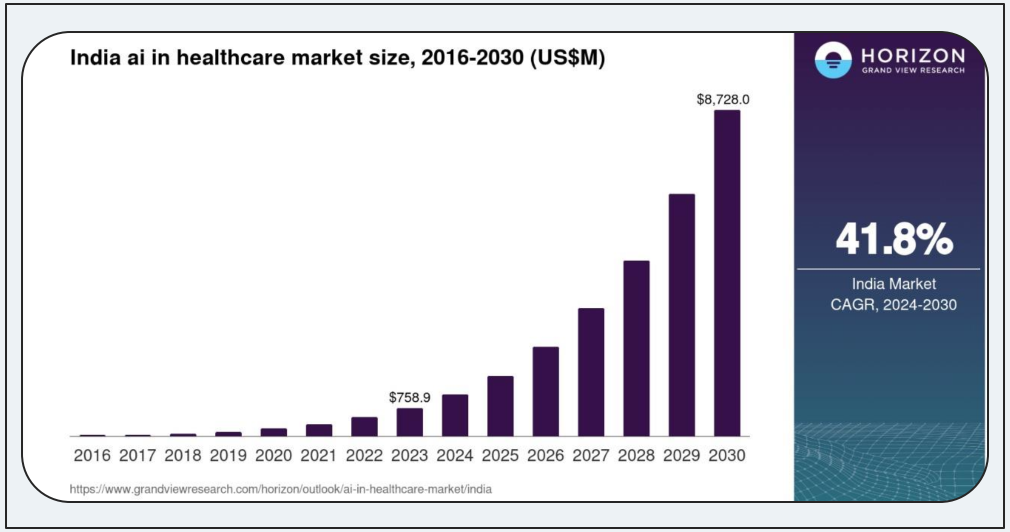
- Patients **lose or struggle** to manage **medical records,** causing repeated tests and delays.
- **Limited access** to doctors leads to **untreated illnesses** and **higher costs**.
- **Over-reliance** on **unverified online** information causes wrong **self-diagnosis**.
- **Rural** and non-English speaking users are excluded due to **language** and **technology barriers.**
- **Emergency care** is **delayed** because nearby **doctors or paramedics** are **hard to locate**.
- **Insurance** processes are **complex**, making healthcare unaffordable for many.
- **Patients** with **low literacy** or typing ability cannot easily communicate their symptoms.
- **Handwritten reports** or **images** are difficult to digitize and share for consultation.



Recent Searches and growth of healthcare sector







MedAI

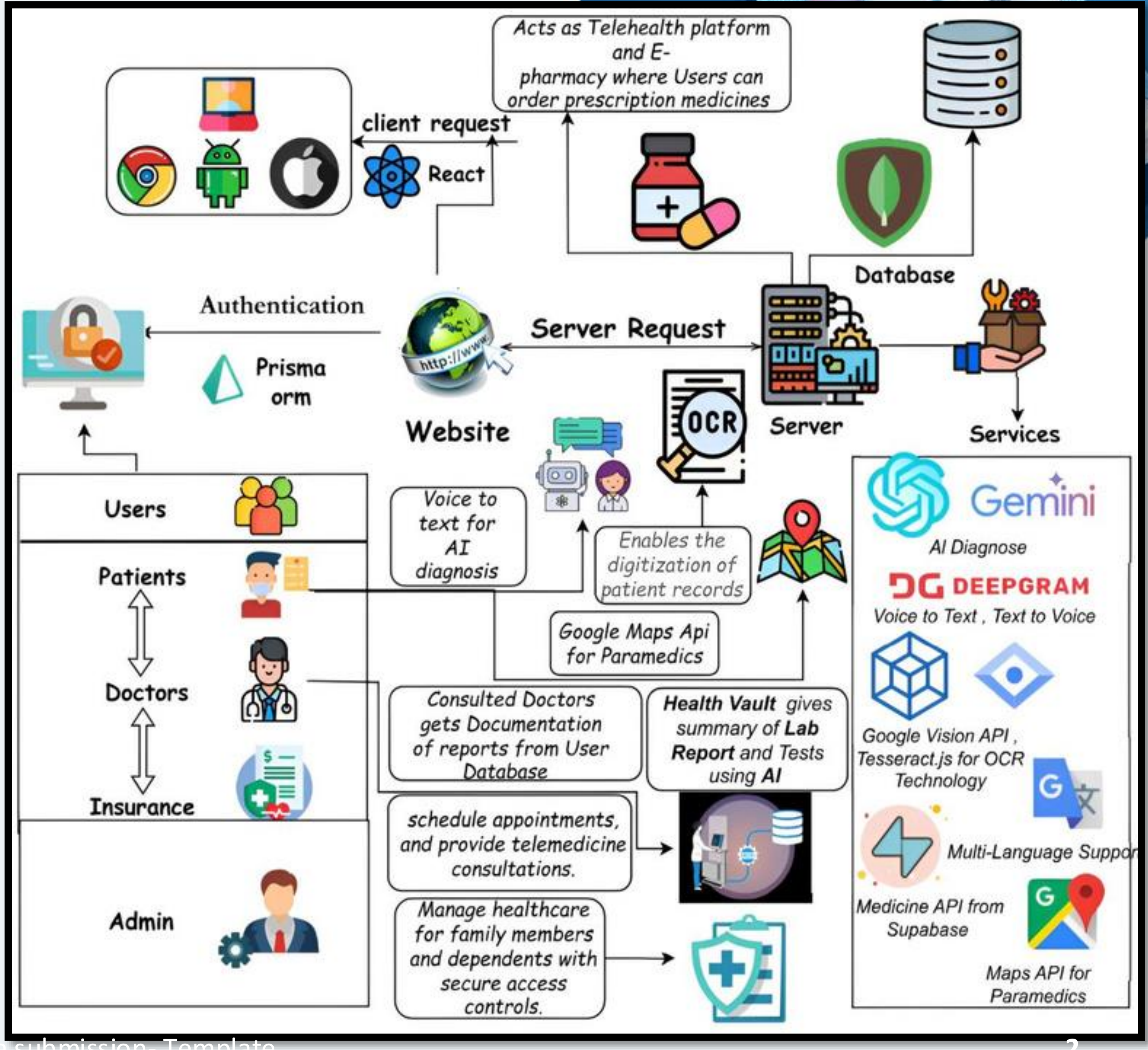
Centralized system for appointments,follow-ups & health records – ensuring continuity of care & smooth patient journey

1)AI Chat Diagnosis – Instant symptom-based guidance, reduces doctor load by filtering common cases, future-ready with AI Avatars for virtual consults.

2) Health Vault – Secure digital storage for reports & prescriptions, upload/share with consent, one-stop health history, paperless & HIPAA/GDPR compliant.

3) Paramedics –Geo-location based first responders, Faster emergency care, strengthens last-mile healthcare delivery.

4) Smart Medicine Search – Usage, dosage & side effects, price comparison, find affordable & correct options.



TECHNICAL APPROACH

WEB DEVELOPMENT

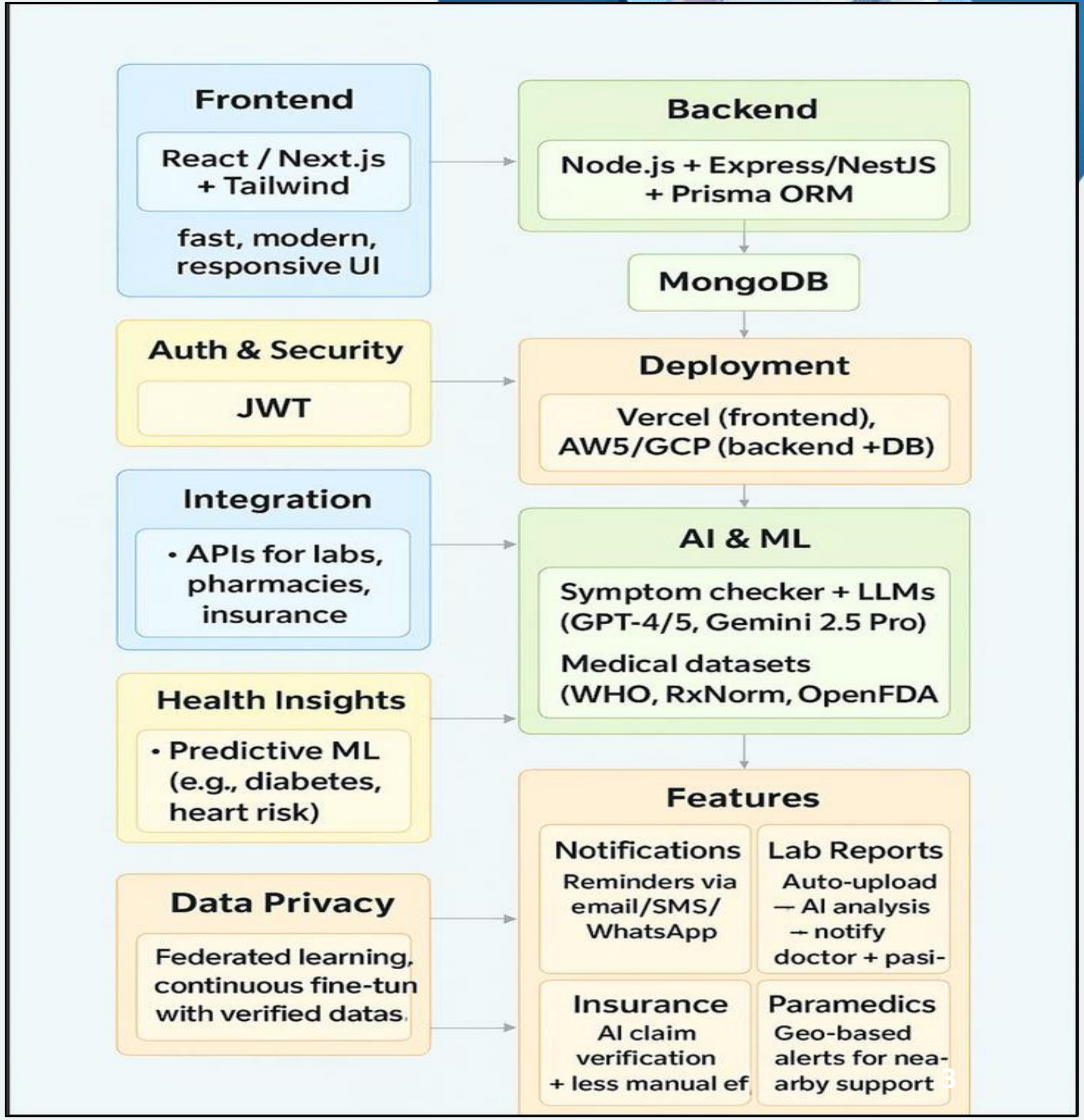
- **Frontend**: React / Next.js + Tailwind for Fast, modern UI
- **Backend**: Node.js(Express / Next JS) + Prisma ORM
- **Database**: MongoDB
- **Auth & Integration**: JWT/Clerk
- **Security**: APIs for Labs, Pharmacies, Insurance

AI & ML

- **Diagnosis**: Symptom Checker + LLMs (GPT-4/5, Gemini 2.5 **RAG**:
- WHO,RxNorm,Open FDA datasets
- **AI Avatar(Future)**:Real-time multimodal agents
- **Health Insights** :Predictive ML (Diabetes, Heart Risk)
- **Data Privacy**:Federated learning + Verified data

FEATURES

- **Notifications**: Email / SMS / WhatsApp reminders
- **Lab Reports**: Auto-upload , AI analysisand Notify doctor & patient
- **Insurance**: AI claim verificationand Less manual effort
- **Paramedics**: Geo-based alerts for nearby support



FEASIBILITY AND VIABILITY

FEASIBILITY

- **Technical** : Proven stack + AI/ML models, HIPAA-ready cloud.
- **Operational** : Scalable portals, easy patient adoption.
- **Economic & Secure** : Low-cost MVP, high ROI, encrypted & compliant.

VIABILITY

- **High Demand** →Rising need for AI triage , digital health records & affordable medicines.
- **Strong Revenue Model** → Subscriptions, commissions & insurance partnerships.
- **Scalable & Adoptable** → ABDM-aligned, easy onboarding, cloud-based nationwide rollout.

CHALLENGES

- **Accuracy & Trust** → Risk of misdiagnosis, user hesitation in AI healthcare.
- **Privacy & Integration** → Strict compliance (HIPAA/GDPR/NDHM), complex system links & high infra costs.

USEUSES

✓ **Patients**→ Symptom-based AI diagnosis, secure health vault, affordable medicine search.

✓ **Doctors & Labs** → Reduced workload, digital record updates, wider patient outreach.

✓ **Paramedics & Insurance** → Faster emergency response, seamless claim verification & fraud reduction.

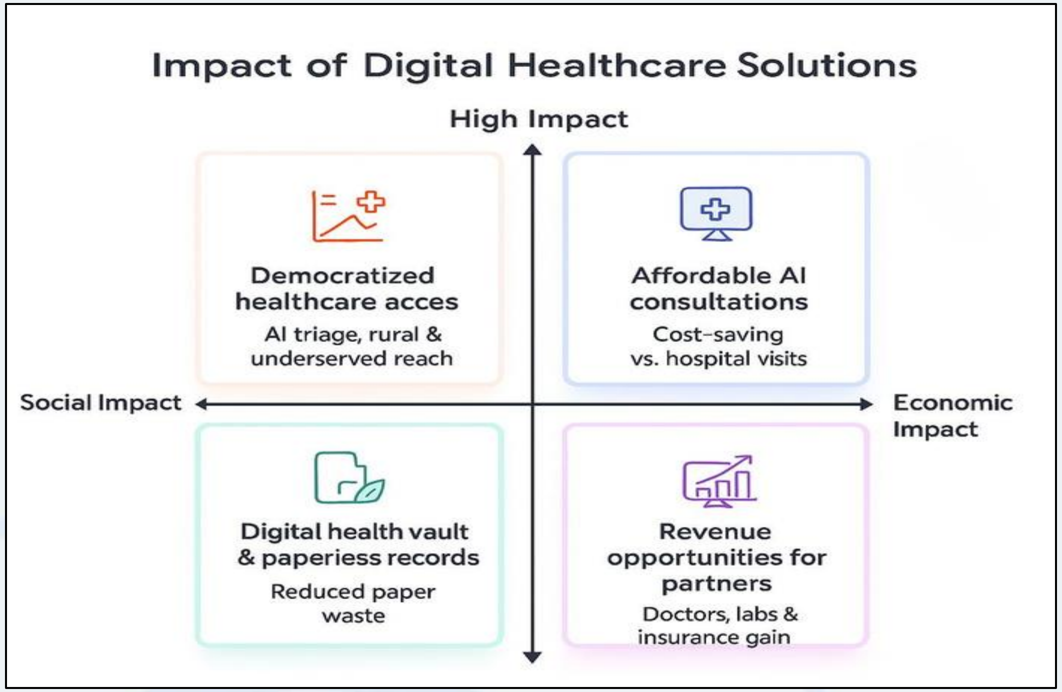
BUSINESS POTENTIAL

✓ **Revenue Streams** → Subscriptions, commissions, insurance tie-ups.

✓ **High Demand** → Growing digital healthcare adoption in India.

✓ **Scalability** → Cloud-based, modular design for nationwide rollout.

IMPACTS AND BENEFITS



Critical Impact on Target Audience

- ❖ **Patients**→ Instant AI triage, secure health vault, afford
- ❖ able medicines.
- **Doctors & Labs**→ Reduced workload, AI-assist for
- ❖ diagnosis , wider outreach.
- **Paramedics & Insurance**→ Faster response, automated claims, fraud reduction.

Benefits of Solution

Social

- Democratizes healthcare → access for rural & underserved populations Builds patient empowerment
- with secure Health Vault Reduces stress on hospitals
- & doctors by handling common cases digitally

Economic

- Affordable AI-driven consultations vs. expensive hospital visits
- Creates revenue opportunities for doctors, labs, and insurance partners
- Enables cost-saving for patients via medicine price comparison

Environmental

- Reduces paper waste through digital records
- Lowers carbon footprint by reducing unnecessary hospital travel.

AI in Medical Diagnosis

NEJM AI (2025) – Performance of GPT-4 in medical diagnosis tasks (nejm.org)

JMIR Formative Research (2025) – AI vs. Physicians in symptom diagnosis accuracy (formative.jmir.org)

Google Research (2025)– Gemini 2.5 Pro capabilities in
multimodal healthcare reasoning (deepmind.google)

Johns Hopkins University (2024) – How Telemedicine is
Redefining Healthcare Access (publichealth.jhu.edu)

HIMSS – Driving the Future of Health with AI (himss.org)

Johns Hopkins Medicine – Telemedicine Startup Connects Patients and Doctors in Rural India (hopkinsmedicine.org)

WHO – Billions left behind on the path to universal health coverage (who.int)

PMC – Reimagining India's National Telemedicine Service to improve access to care -

Parameter	Competitors (Practo, MFine, Qure.ai, Ada, Babylon, etc.)	How MedAI Ticks This Box
AI Chat Diagnosis	Practo: mostly human doctors; MFine, Ada: limited AI triage; Babylon: faced accuracy scrutiny.	Full AI diagnosis via chatbot → AI Avatars, using GPT-4/5, Gemini, RAG for context accuracy.
Health Vault (Records)	Practo : limited EHR; ABDM: Digi Locker/ ABHA vaults; Apple Health /NHS: used abroad.	Secure Health Vault: consent-driven, ABDM/NDHM & HIPAA/GDPR compliant, interoperable with labs & hospitals.
Paramedic / Emergency Access	Competitors: mostly ignore this, except ambulance apps like Dial4242.	Geo-tagged Paramedic integration for nearby first responders, linked with patient vault for allergies/history

Demo Link: (https://youtu.be/56WfC0BVt7Y)